ORAL REHYDRATION SALTS

First-line for dehydration

Mix with clean water

CHEMICAL NAME Glucose anhydrous 13.5 g/L + sodium chloride 2.6 g/L + potassium chloride 1.5 g/L + trisodium citrate 2.9 g/L (WHO low-osmolarity ORS: 245 mOsm/L)

CLASS Oral electrolyte and glucose replacement solution

TRADE NAMES WHO-ORS · Dioralyte · Pedialyte · Rehydran

THERAPEUTIC USES / INDICATIONS

- Prevention and treatment of dehydration from acute diarrhoea (children and adults), including cholera.
- Mild–moderate dehydration from vomiting, heat illness.
- Maintenance during gastroenteritis (with zinc in children).

ADMINISTRATION

- **PO:** dissolve sachet in the exact volume of clean (boiled and cooled) water — do not add sugar.
- Give small, frequent sips (spoon, cup); for vomiting child: 5 mL every 1–2 min.
- Discard mixed solution after 24 h.
- **NG tube** if unable to drink but not shocked.

DOSAGES

- **Some dehydration (child):** 75 mL/kg over 4 h, then reassess.
- **After each loose stool (by weight):** <5 kg 50 mL; 5–<10 kg 100 mL; 10–20 kg 200 mL; >20 kg 300 mL.
- **Adults:** replace ongoing losses; drink as much as wanted (sachet label volumes).
- Continue breastfeeding and normal feeding.

PHARMACOKINETICS

- Glucose–sodium co-transport (SGLT1) in small intestine remains intact in secretory diarrhoea.
- Absorbed water and electrolytes restore circulating volume within hours.

PHARMACODYNAMICS

- **Sodium–glucose co-transport** drives Na^+ and water absorption across gut wall.
- Citrate corrects metabolic acidosis; K^+ replaces losses.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Vomiting if drunk too quickly.
- Salty taste.

ADVERSE EFFECTS (SERIOUS)

- Hypernatraemia if made up too concentrated.
- Fluid overload (renal/cardiac failure) — rare.

CONTRAINDICATIONS & CAUTIONS

- Severe dehydration with shock (needs IV fluids).
- Intestinal obstruction, paralytic ileus.
- Persistent vomiting preventing intake; severe renal failure (caution).

KEY INTERACTIONS

- None significant.

NURSING MANAGEMENT

- Assess dehydration (WHO signs: thirst, sunken eyes, skin pinch, lethargy) and weight.
- Teach correct preparation to caregivers.
- Record intake, stool output, urine output.
- Give zinc to children with diarrhoea; continue feeding.
- Escalate to IV if severe or not improving.

PATIENT EDUCATION

- Mix one sachet with exactly the stated volume of clean water.
- Give small sips often; if child vomits, wait 10 min then continue slowly.
- Keep breastfeeding.
- Return if child is very sleepy, not drinking, has blood in stool or few wet nappies.

KEY POINTS

- Glucose–sodium co-transport rehydration.
- Exact water volume; no added sugar.
- 75 mL/kg over 4 h for some dehydration.
- Plus zinc and continued feeding in children.

SOURCES CHECKED **[563]** Dioralyte Blackcurrant Sachets (UK SmPC) **[564]** MSF clinical guidelines (appendices) **[565]** WHO diarrhoeal disease fact sheet

Full references p. 432–457